Qingyang Zhou

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https://esperarr.github.io/CV/

EDUCATION

Tsinghua University September 2022 - Present
Xinya College, Tsinghua University: https://www.xyc.tsinghua.edu.cn/index.htm
Department of Computer Science and Technology, Tsinghua University: https://www.cs.tsinghua.edu.cn/
GPA 3.6/4.0
Selected A/A+ coursework: Complex Analysis (A+), Media Computing(A+), Database Special Topic Training(A+), Introduction to Artificial Intelligence, Introduction to Machine Learning, Fundamentals of Computer Graphics, Students Research Training, Programing and Training.

RESEARCH PROJECTS

"Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation"
Yuwei Zhou, Xin Wang, Hong Chen, Yipeng Zhang, ***Qingyang Zhou****, Wenwu Zhu*
OpenReview Paper PDF

"Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion"
Zipeng Zhang , Shiwei Wu, Ran Ran, ***Qingyang Zhou****, Ranrui Ma, Jiaye Leng, Jian Zeng, Zhenhui, Chun Yu*
Paper PDF

INTERNSHIP / RESEARCH

Client Development Engineer Intern at Tencent (Project MAX) May 2026 - August 2026
Tencent, Project MAX Team

- Optimized NavMesh pathfinding to improve agent movement smoothness, resolving instability on terrain bumps and convex corner edges.
- Implemented NPC auto-navigation and pathfinding behavior using custom FixNavMesh algorithms.
- Developed a GM panel for debugging and performance stats toggling in the battle UI.
- Built and refined game control panel UI for the 2V2 battle mode.

Scientific Research Intern at Multimedia and Network Big Data Lab July 2023 - July 2024
Tsinghua University Multimedia and Network Big Data Lab: mn.cs.tsinghua.edu.cn

- Contributed to the research project "Video Generation System Combining Large Language Models with Diffusion Models."
- Studied diffusion models and personalized image generation.
- Assisted with the research and manuscript preparation for "Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation."

Scientific Research Intern at Pervasive Human-Computer Interaction Lab January 2025 - May 2025
Tsinghua University Pervasive Human-Computer Interaction Laboratory: pi.cs.tsinghua.edu.cn

- Conducted research on aligning human and AI thought processes.
- Developed a human-AI interactive writing assistant to support collaborative idea generation.
- Contributed to the research and manuscript preparation for "Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion."

Scientific Research Intern at Information Retrieval Lab October 2024 - July 2025
Information Retrieval Lab at Tsinghua University: thuir.cn

- Focused on improving traditional recommendation models (such as FM, DeepFM, AFM).
- Incorporated SAE plugins into these models to enhance their interpretability, controllability, and transparency.

POSITIONS OF RESPONSIBILITY

Volunteer Teaching Assistant (Programming and Algorithms) September 2025 - Present
Tsinghua University, Center for Student Learning and Development: learning.tsinghua.edu.cn

- Provided drop-in tutoring for programming and computer science courses, including C/C++ programming, data structures, and algorithms.
- Assisted students with debugging, algorithm design, and understanding core programming concepts during regular tutoring sessions.

Secretary of the Intelligent Agent Department of the Student Science and Technology Association February 2023 - Present
Tsinghua University Computer Science and Technology Department, Student Science and Technology Association, Intelligent Agent Department: net9.org/home

- Participated in the game development and event organization for the Intelligent Agent Competition.
- Mainly responsible for the development of the Intelligent Agent-Based Game 'Generals' Project in the 28th Tsinghua University Intelligent Agent Competition.

Tutor of Tsinghua University Science and Engineering Summer Camp July 2024 - August 2024
Tsinghua University 2024 Science and Engineering Summer Camp, I6 Class Tutor

- Organized engineering activities and mentored students through intelligent robot design and card-bridge construction.

SKILLS

Languages	Chinese (native), English (fluent), Spanish (basic)
Programming	C/C++, Python, Java, JavaScript/TypeScript, Rust, Verilog/SystemVerilog
Graphics / Games	OpenGL, Unity, path tracing, rendering pipelines
AI / Machine Learning	PyTorch, NumPy, machine learning and deep learning fundamentals
Tools	Git, Linux, Docker, LaTeX, Markdown

PROJECTS

Intelligent Agent-Based Game "Generals" September 2023 - May 2024
Major Project in the 28th Tsinghua University Intelligent Agent Competition

- A turn-based strategy game available at saiblo.net/game/35.
- Responsible for backend logic, unit tests, debugging, SDK development, documentation, and guiding participants through the competition.

Path Tracing Renderer [Source code] June 2026
Computer Graphics Course Project

- A physically-based path tracer supporting both Whitted-style ray tracing and Monte Carlo path tracing with Russian Roulette for unbiased termination.
- Implemented PBR materials with GGX microfacet importance sampling (isotropic and anisotropic), Cook-Torrance BRDF, and Fresnel Schlick, covering diffuse, mirror, glass, and glossy surfaces.
- Added BVH acceleration for 100k+ face meshes, NEE + MIS with power heuristic weighting, depth of field, MSAA anti-aliasing, gamma correction, and OpenMP parallelization.

Text Sentiment Analysis Model [Source code] May 2024
Course Project

- Built CNN, RNN, and MLP sentiment classifiers in PyTorch and evaluated their performance, reaching over 90% accuracy.

Five-Stage Pipeline CPU [Source code] November 2025 - December 2025
Computer Organization Course Project

- Implemented a five-stage pipeline CPU in SystemVerilog for 19 basic RISC-V instructions as well as some extensions, including forwarding, signal control, and hazard handling; validated with Vivado simulation.

Connect Four AI [Source code] June 2024
Course Project

- Developed a Python Connect Four AI using Monte Carlo Tree Search (MCTS) with Upper Confidence bounds for Trees (UCT), achieving over 90% win rate against baselines.
- Implemented UCT-based node selection and simulation strategies for efficient game-tree exploration.